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To cite this article: Yongsuk Yun & Hyun Ah Kim (2022) High investor sentiment and initial engagement discounts: evidence from Korea, *Applied Economics*, 54:8, 959-973, DOI: [10.1080/00036846.2021.1971619](https://doi.org/10.1080/00036846.2021.1971619)

To link to this article: <https://doi.org/10.1080/00036846.2021.1971619>



Published online: 08 Sep 2021.



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High investor sentiment and initial engagement discounts: evidence from Korea

Yongsuk Yun^a and Hyun Ah Kim^b

^aDepartment of Accounting, Hannam University, Daejeon, Korea; ^bSchool of Industrial Management, Korea University of Technology & Education, Cheonan City, Chungnam Province, Korea

ABSTRACT

This study examines whether high sentiment affects auditors' risk assessment in a less litigious environment. Specifically, we use a Korean setting with relatively low litigation risks and investigate whether audit fees are lower when firms change auditors during high sentiment times. Lower audit fees responding to high sentiment in a weak legal environment may indicate that auditors perceive the overall audit risk including reputation risk and regulation risk as low within the given period, increasing the possibility of their failure to detect managerial opportunistic behaviour. Using 17,469 Korean public firm-year observations from 2003 to 2017, we find that initial audit fees are significantly lower when investor sentiment is high, while changing auditors has no influence on audit fees when investor sentiment is low. Furthermore, this result is maintained even when auditors are changed to Big4 or industry specialist auditors (SPEC), which are known to have high ability and risk sensitivity. We also find that Big4 and SPEC initial auditors input significantly less audit hours during high investor sentiment periods, suggesting that optimistic bias caused by high sentiment makes auditors expect that relaxed audits are tolerable regardless of the legal environment.

KEYWORDS

High investor sentiment; initial engagement discounts; audit fees; audit hours; auditor change type

JEL CLASSIFICATION
M42; G14

1. Introduction

We analyse the effect of investor sentiment on external auditors' risk assessments, measured by initial engagement discounts. The accounting literature assumes that investors are rational, but recent studies show that investor sentiment,¹ a cognitive bias, can cause deviations from the fundamental values of stock prices (Baker and Wurgler 2006). Social recognition argues that optimistic individuals pay limited attention to information, while pessimistic individuals make a thorough investigation (Bless et al. 1996). Consistent with this, investors in high sentiment periods respond more to unexpected positive earnings and over-value stock prices (Baker and Wurgler 2006; Seok, Cho, and Ryu 2019). Managers and analysts opportunistically use this situation by providing optimistic forecasts during high sentiment times (Hurwitz 2018; Walther and Willis 2013). Given that investors use accounting variables in their investment decisions and earnings are value relevant in high

sentiment periods (He, Hong, and Wu 2020), auditors' role as an external monitor may be important in reducing investors' biases.

Auditors should identify and assess the risk of material misstatements when performing risk assessment procedures (Public Company Accounting Oversight Board (PCAOB) 2010). The perceived audit risk, based on predicted litigation, reputation loss, or regulation intervention, increases the audit scope and the number of audit hours, which acts as an important determinant of audit fees (DeFond and Zhang 2014).² In addition, auditors require premiums responding to the risk that cannot be eliminated completely with auditor effort, which also results in higher audit fees (DeFond and Zhang 2014). High sentiment may be a potential audit risk because it provides managers with opportunities to manage earnings, which hinders investors' proper decision-making through profit inflation (Ali and Gurun 2009; Amin, Eshleman, and Guo 2021a; Simpson 2013).

CONTACT Hyun Ah Kim  kukolo@koreatech.ac.kr  School of Industrial Management, 1600 Chungjeol-ro, Byeongcheon-myeon, Dongnam-gu, Cheonan City, Chungnam Province 31253, Korea

¹Investor sentiment is defined 'as a belief about future cash flows and investment risks that is not justified by the facts at hand,' and indicates investor optimism or pessimism (Baker and Wurgler 2007).

²'Litigation risk exposes auditors to financial penalties, while reputation risk impairs the ability to attract and retain clients. Regulation risk is the threat of regulatory intervention, which subjects auditors to sanctions that include fines and criminal penalties.' (DeFond and Zhang 2014, 296).

Amin, Eshleman, and Guo (2021b) also support this by showing that auditor change from Big4 to non-Big4 for better internal control opinion is prevalent in high sentiment times, which results in greater restatements.

If auditors recognize that high sentiment is associated with the distortion of financial statements leading to future lawsuits, they will increase the scope of the audit and spend more audit hours to ensure that financial statements are free of material misstatements (professional duty view). However, because high sentiment attracts investors to be insensitive to financial statements, auditors are likely to reduce their audit effort in the belief that litigation risk due to financial misreporting will be low in high sentiment times (commercial incentives). Amin, Eshleman, and Guo (2021a) examine two contrasting views, auditors' professional duties and commercial incentives,³ and find that audit fees are lower on average in high sentiment periods, consistent with commercial incentives.

The findings of prior studies are based on the US setting, where institutional investors are dominant in securities markets and the legal environment is strong; thus, high litigation risk sensitivity can affect auditor behaviour. Therefore, the effect of high sentiment on auditors needs to be examined by using different legal environments. In this study, we use a Korean setting with less developed stock markets and a weak legal environment. This is important because it can offer regulatory policy implications for auditors in less litigious countries. Although the most serious risk of threatening auditors is litigation risk, it is not clear whether the legal system affects prior findings because auditors are influenced by the state of their mood (Cianci and Bierstaker 2009).

The Korean stock market is 'characterized by significant information asymmetry among investor types, high market sentiment, and prevalent investor psychology' (Ryu, Kim, and Yang 2017, 826). The litigation against auditors has increased since class action lawsuits became effective in 2005, but the litigation risk is relatively low compared to the

US, considering that 18 of the 71 lawsuits that ended from 2016 to 2018 ordered audit firms to pay damages totalling 20.1 billion won (about \$ 2 million) (Financial Supervisory Service (FSS) 2019). Therefore, Korea provides an appropriate setting to examine whether the prior finding is affected by the litigation environment. If low perception of audit risk in high sentiment times is maintained regardless of the litigious environment, it may indicate auditors' vulnerability to high sentiment, failing to correct investors' biases.

In particular, we focus on auditor changes, creating an environment where management can exert pressure (Huang, Raghunandan, and Rama 2009). The results of Amin, Eshleman, and Guo (2021b) show that high audit fees for risky firms, which change auditors for favourable internal control opinion in high sentiment periods, do not significantly reduce the likelihood of restatements. Practitioners and academics are interested in the existence of initial discounts because it can lead to less independent auditors (Cohen Commission 1978; DeAngelo 1981; Dye 1991). Korean public or sizable companies are required to be audited by the External Audit Act, so they do not value audits highly, which leads to low audit fees on average.⁴ In this environment, lower audit fees decrease auditors' incentives to input more effort in auditing, increasing potential audit failure. Therefore, if auditor changes in high sentiment periods are related to lower audit fees, potential problems are likely to be poorly identified and monitored by auditors.

Using 17,469 firm-year observations of Korean listed firms from 2003 to 2017, we find that initial audit fees are significantly lower when investor sentiment is high, while auditor change has no influence on audit fees in low investor sentiment periods. Furthermore, this result is maintained even when auditors are changed to Big4 or industry specialist auditors (SPEC), which are known to have high ability and risk sensitivity. This shows that high sentiment influences auditor behaviour in a less litigious environment, implying that the

³Professional duty view argues that auditors spend more audit hours to reduce audit risks to an acceptable level in high sentiment periods, which leads to higher audit fees. In contrast, commercial incentives view suggests that auditors predict lower litigation risk due to investor inattention in high sentiment times, resulting in lower audit fees (Amin, Eshleman, and Guo 2021a).

⁴Korean audit fees are about 90%, 70%, 40% lower than those of the US, Japan, and China, respectively (Kwon and Kim 2019). Korean government has made efforts such as establishing standard for appropriate audit hours to ensure auditors' independence and to improve audit quality (Korea Herald 2018).

auditor's governance responsibilities do not work properly when sentiment is high. Auditor change is a negative event that investors look into because it increases information risk (Francis et al. 2017; Han et al. 2021). However, optimistic bias caused by high sentiment makes them generous to it, enabling auditors to accept lower initial audit fees with the expectation that a relaxed audit is tolerable. Second, Big4 or SPEC initial auditors input significantly less audit hours in high investor sentiment, suggesting that auditors reduce audit effort with the judgement of low audit risk. Given the positive association between high sentiment and earnings management (Ali and Gurun 2009; Simpson 2013), these findings show that auditors' assessment of low audit risk in high sentiment is more likely to exacerbate investors' biases rather than lessen them because loose audits are difficult to detect and correct managerial opportunistic behaviour.

For the robustness of our results, we conduct an additional analysis considering the types of auditor change such as Big4 to Big4, Big4 to non-Big4, non-Big4 to Big 4, and non-Big4 to non-Big4. The results of the additional test are consistent with our main findings except for non-Big4 to non-Big4. Given that the audit fee of Big4 is much more than that of non-Big4, the evidence that initial engagement discounts exist in auditor change from non-Big4 to Big4 is somewhat surprising. This may indicate a substantial effect of investor sentiment on the risk assessment and audit pricing of auditors.

This study contributes to the literature in several ways. First, we extend the literature on investor sentiment by showing that prior findings are not sensitive to the legal environment. Our results of accepting lower audit fees even in auditor change, which requires more audit efforts to obtain reasonable assurance about financial statements due to information risk, indicate that the monitoring system is not working well in high sentiment periods, raising the need for regulators to take various measures to strengthen their monitoring mechanisms to reduce the possible negative impact of high investor sentiment.

Second, we use a firm-level investor sentiment variable, unlike earlier studies that adopt a time-level measure. It is appropriate for Korean stock

markets that are dominated by individuals (Ryu, Kim, and Yang 2017; Seok, Cho, and Ryu 2019). It also overcomes the limitation that investor sentiment is market-wide despite being affected by various factors, and enables researchers to consider individual firms' differences in their studies. Our evidence extends prior results at an auditor level (Chung, Cohen, and Monroe 2008) and a time level (Amin, Eshleman, and Guo 2021a) to a firm level and enhances the understanding about the influence of cognitive biases on auditor risk assessment. Prior psychological research on auditor decisions focuses on experimental settings (Chung, Cohen, and Monroe 2008; Cianci and Bierstaker 2009), but the current study contributes to the literature by adding empirical results.

Third, we provide a new factor affecting initial engagement discounts. Prior research has found the cause of fee discounting in audit-related characteristics, such as changes in audit firms or audit partners (Huang et al. 2015). This study shows that high sentiment, a non-audit characteristic, can also affect audit risk assessment, thereby lowering initial audit fees. Our results suggest that initial audit fees should be investigated from various perspectives that affect managerial or auditor decision-making.

The remainder of this study is organized as follows. Section 2 reviews prior studies and develops the hypotheses. Sections 3 and 4 present the research methodology and empirical results, respectively. Finally, Section 5 concludes the paper.

II. Literature review and hypothesis development

Prior research on initial audit fee discount

Initial discounting is an important issue for practitioners and academics in the auditing field because it can impair auditors' independence or audit quality for continuous future contracts by lowering audit hours in response to decreased audit fees (Cohen Commission 1978; DeAngelo 1981; Dye 1991; Eshleman and Guo 2014). Initial auditors may discount their audit fees to satisfy new clients expecting lower audit fees or to win in price competition (Beattie and Fearnley 1995). DeAngelo (1981) argues that it is a sunk cost that does not affect the independence of auditors because it is 'a

competitive response to the expectation of future quasi-rents to incumbent auditors'. However, Simon and Francis (1988) mention that initial discounting plays a role in impairing audit quality based on prospect theory suggesting that people give more value to gains than losses. Consistent with this, Public Company Accounting Oversight Board (PCAOB) (2011, 17) states, 'If auditors bid on new engagements with the assumption that they will lose money in the first years of an engagement but recoup that loss over a long period of time, the problem may be unrealistic pricing, with a resulting effect on audit effort or resources at the beginning of an auditor-client relationship'. Turner (2002) and Hansen (2010) show concerns over the possibility that low audit fees may lead to deterioration in audit quality.

Empirical evidence is mixed and differs depending on the country or time. Specifically, studies examining the US audit market report the existence of an initial discounting behaviour. For example, Simon and Francis (1988) find that initial audit fees are lower by about 24% in the first year and 15% in the next two years, and Ettredge and Greenberg (1990) report a 24% discount. Ghosh and Lustgarten (2006) are motivated by the different competitions in small and large audit firms and show that fee discounting is more prevalent in non-Big4 than Big4. Recent studies have analysed the existence of initial discounting in the pre- and post-SOX periods, but the results are still mixed. Huang, Raghunandan, and Rama (2009) report the initial audit fee premium for Big4 auditors in the post-SOX period, but Desir, Casterella, and Kokina (2014) find that initial fee discounting exists in both Big4 and non-Big4 from 2007 to 2010, in the post-SOX period. In the UK audit market, prior studies provide relatively consistent evidence on the existence of initial discounting (Gregory and Collier 1996; Peel 2013), but the empirical results of the Australian audit market are mixed (Francis 1984; Craswell and Francis 1999). Regarding the Korean audit market, prior studies provide evidence that auditors discount their initial fees (Rho, Bae, and Cho 2004; Lee, Park, and Jeong 2011; Jeong and Ki 2016). In addition, Shin, Choi, and Cho (2007)

find a negative association between initial discounts and initial audit quality, supporting the concern raised by regulators or academics.

Prior research on investor sentiment and financial reporting

The literature has analysed the effect of investor sentiment on corporate decisions (e.g. Grundy and Li 2010), but the accounting field has just started to examine its influence on managers and analysts. In terms of managers, Hurwitz (2018) reports that high investor sentiment is associated with optimistic management forecasts and interprets the result as evidence of managerial vulnerability to investor sentiment. Consistent with this, Brown et al. (2012) find that pro forma earnings disclosures tend to exclude recurring and non-recurring expenses in high investor sentiment. Bergman and Roychowdhury (2008) show that managers are less likely to disclose long-horizon earnings forecasts in high investor sentiment because they want to maintain investor optimism.

Furthermore, prior studies examining the effect of investor sentiment on analysts' behaviour provide evidence that analysts are also susceptible to high investor sentiment. Specifically, Hribar and McNinnis (2012) find that analysts forecast the long-term earnings of firms with high uncertainty more optimistically in high investor sentiment periods. Walther and Willis (2013) use the short-run quarterly earnings forecast of sell-side analysts and find that they tend to forecast more optimistically and less accurately when investor sentiment is high.

Several studies investigate the relationship between investor sentiment and financial reporting such as accruals or conservatism. Ali and Gurun (2009) examine the association between investor sentiment and accruals anomaly and find greater accruals for small firms in high sentiment periods, supporting investors' limited attention to accruals and managers' opportunistic usage. Simpson (2013) extends this study by adding evidence that managers use accruals to meet or beat analysts' forecasts during high investor sentiment periods. Contrary to previous results, Ge, Seybert, and Zhang (2019) show that managers are likely to report conservatively in high sentiment times to reduce potential litigation risk due to stock price decline.

Hypothesis development

Extending the literature to auditors' risk assessment, Amin, Eshleman, and Guo (2021a) investigate external auditors' reactions to investor sentiment using the US data and find that audit fees are lower in high sentiment periods despite the high likelihood of misstatements. They additionally show that market reactions to misstatements are less negative in high sentiment periods and argue that a decrease in auditors' legal risk leads to lower audit fees.

Audit fees consist of audit effort costs and expected liability losses arising from audited financial statements (Simunic 1980; Seetharaman, Gul, and Lynn 2002). The positive association between litigation risk and audit fees in prior studies indicates that it plays an important role in determining audit fees (Seetharaman, Gul, and Lynn 2002; Choi et al. 2009). Countries have different legal liabilities depending on their legal regimes, which vary the degree of litigation liability reflected in audit fees (Choi et al. 2009). Therefore, if the prior finding based on the US which has a strong legal environment is mainly related to litigation risk, the result is unlikely to be found in countries with weak legal regimes.

However, engagement risk includes reputation and regulation risks as well as litigation risk (DeFond and Zhang 2014). Auditors in less litigious environments do not expect a significant reduction in litigation risk from high sentiment, but investors' positive biases can entice auditors to conduct less conservative audits, expecting investors to be tolerant of financial statement errors. This may decrease audit fees by lowering auditors' reputation risk.

In addition, auditors are affected by mood states (Chung, Cohen, and Monroe 2008; Cianci and Bierstaker 2009). Chung, Cohen, and Monroe (2008) use the experiment with Australian auditing students and find that a positive mood is associated with less conservative inventory valuation. Cianci and Bierstaker (2009) show that auditors in the experiment are good at hypothesis generation tasks, such as cause analysis of fluctuations under a negative mood. Even if investors' high sentiment may not influence auditors' mood directly, there is a possibility that the favourable atmosphere

surrounding a firm in the stock market can affect auditors' engagement risk evaluation, thereby reducing audit fees.

To examine the effect of high sentiment on auditor risk assessment in Korea with relatively low litigation risk, we focus on auditor change. With all other things equal, it increases audit engagement risk because of high information risk derived from the lack of knowledge about a newly accepted auditee (Francis et al. 2017). If the influence of high sentiment is not limited to litigation risk, audit fees are expected to be lower when firms change auditors. However, auditors require higher audit fees in response to future misstatements of firms that engage in audit opinion shopping during high sentiment times (Amin, Eshleman, and Guo 2021b). Therefore, it is difficult to predict the effect of high sentiment on initial audit fees in a weak legal environment. Thus, we set our hypothesis as follows.

Hypothesis. High investor sentiment has no influence on initial audit fees in a less litigious country.

III. Research design

Individual firm-level investor sentiment measure

In this study, we use firm-level investor sentiment to examine its effect on initial audit fees. Specifically, we measure the variable based on Ryu, Kim, and Yang (2017) to fit the Korean stock market. Baker and Wurgler (2006) suggest a market-based index of sentiment using six proxies: the closed-end fund discount rate, NYSE stock turnover, the number of IPOs, average first-day returns on IPOs, the share of equity issues, and the dividend premium. However, it is difficult to apply this method to Korean studies. This is because: (1) unlike the US stock market, the Korean stock market has a small margin of rate depending on dividends; thus, we cannot use the dividend premium; (2) the number of IPOs in the Korean stock market is significantly less than that in the US, so it is not appropriate to use IPO-related variables; (3) there are insufficient data on the discount rate for closed-end funds. Therefore, prior Korean studies claim that it is appropriate to

apply the said method to the Korean market (Kim and Byun 2010; Ryu, Kim, and Yang 2017). Moreover, in our study, we focus on the sentiment of individual companies, unlike most previous studies, so we measure the individual firm-level sentiment variable based on Ryu, Kim, and Yang (2017). It is calculated at an individual stock level; thus, it is known to better capture firm-level characteristics.

To construct a comprehensive individual firm-level sentiment index, we adopt principal component analysis to extract the index explaining investor sentiment based on the following five daily measures that capture investor sentiment and market expectations: relative strength index (RSI), psychological line index (PLI), adjusted turnover rate (ATR), logarithm of trading volume (LTV), and individual investor buy-sell imbalance (IBSI). The method of measuring the sentiment index is a process for extracting a common part with various variables that include information about the sentiment. Thus, these five measures can be related, and the purpose of principal component analysis is to extract only the common parts of these related variables.

Relative strength index

RSI captures whether the stock is oversold (20 or below) or overbought (80 and above) by calculating the percentage of increase in stock price over a certain period. We use the 14-day when calculating RSI as in Chong and Ng (2008) and Ryu, Kim, and Yang (2017). Specifically, RSI is calculated as follows:

$$RSI_{i,t} = \frac{RS_{i,t}}{1 + RS_{i,t}} \times 100$$

where R

$$S_{i,t} = \sum_{n=0}^{13} \max(P_{i,t-n} - P_{i,t-1-n}, 0) / \sum_{n=0}^{13} \max(P_{i,t-1-n} - P_{i,t-n}, 0) \quad (1)$$

$P_{i,t}$: the closing price of stock i at time t .

Psychological line index. PLI indicates the psychological stability of investors and measures whether the stock is oversold (25 or below) or overbought (75 and above) by calculating the rising and falling days of stock price for a certain period, typically 12 days (Yang and Gao 2014). Both RSI and PLI are indicators that reflecting the investor

sentiment by measuring investors' overheated sentiment or stagnant sentiment on the stock and have a scale from 0 to 100. PLI is calculated as follows:

$$PLI_{i,t} = 100 \times \sum_{n=0}^{11} \left(\frac{\max(P_{i,t-n} - P_{i,t-1-n}, 0)}{P_{i,t-n} - P_{i,t-1-n}} \right) \quad (2)$$

$P_{i,t}$: the closing price of stock i at time t .

Adjusted turnover rate

ATR determining whether investors are optimistic or pessimistic relies on the positive relation between liquidity and investor sentiment (Baker and Stein 2004; Kim and Byun 2010; Yang and Zhang 2014). The positive ATR under positive stock returns represents a bullish market state. ATR is calculated as follows:

$$\begin{aligned} ATR_{i,t} &= \frac{Volume_{i,t}}{n \text{ of shares outstanding}_{i,t}} \\ &\quad \times \frac{R_{i,t}}{|R_{i,t}|}, \text{ where } R_{i,t} \\ &= \frac{P_{i,t}}{P_{i,t-1}} - 1 \end{aligned} \quad (3)$$

$Volume_{i,t}$: the trading volume of stock i at time t
 $n \text{ of shares outstanding}_{i,t}$: the number of shares outstanding of stock i at time t .

Logarithm of trading volume

LTV is a similar measure with ATV to measure investor sentiment coined in Baker and Stein (2004) and Liao, Huang, and Wu (2011). LTV is the natural log number of daily trading volume of each stock as $LTV_{i,t} = \ln(Volume_{i,t})$.

Individual investor buy-sell imbalance

IBSI indicates trading behaviour among retail investors using buy sell volume of stock (Kumar and Lee 2006; Kim and Park 2015). Positive IBSI value represents the existence of net demand for the stock, which means that investor sentiment is increasing for the periods (Chiang, Tsai, and Lee 2011). IBSI is calculated as follows:

$$IBSI_{i,t} = (BV_{i,t} - SV_{i,t}) / (BV_{i,t} + SV_{i,t}) \quad (4)$$

$BV_{i,t}$ and $SV_{i,t}$: the buy and sell volume of individual investor for stock i at time t .

Based on the above five proxies, we construct a composite index using a principal component analysis to isolate the common sentiment component. It is a widely used procedure to measure investor sentiment in the literature (e.g. Baker and Wurgler 2006; Chen, Chong, and Duan 2010). Raw values of investor sentiment index ($S_{i,t}$) are calculated for each firm by multiplying the first principal component by the corresponding variable:

$$S_{i,t} = F_{i,RSI} \times RSI_{i,t} + F_{i,PLI} \times PLI_{i,t} + F_{i,ATR} \times ATR_{i,t} + F_{i,LTV} \times LTV_{i,t} + F_{i,IBSI} \times IBSI_{i,t} \quad (5)$$

The raw values of the investor sentiment index ($S_{i,t}$) in equation (5) has a problem in that it cannot distinguish common sentiment components from common market factors. Because our sentiment index focuses on representing individual firm-level investor sentiment, we need to control the overall market influence from the investor sentiment index ($S_{i,t}$). Thus, to eliminate the possible effects of the market-wide return component we perform the following regression based on equation (6) and orthogonalize the sentiment index proxies ($S_{i,t}$) with respect to market excess returns.

$$S_{i,t} = \alpha_0 + \alpha_1 \times MKTEXR_t + \varepsilon_{i,t} \quad (6)$$

$MKTEXR_t$: the excess returns of the overall stock market, the risk-free rate used in is the CD91 rate (91-day certificate of deposit rates).

Finally, the orthogonalized component, the residual ($\varepsilon_{i,t}$) in equation (6), is the firm-level investor sentiment measure for individual stock i . In our analysis, we use its annualized value as the main variable ($Sentiment_{i,t}$). Depending on the sign of the variable, we divide our sample into two groups; SENTIMENT HIGH (positive $Sentiment_{i,t}$) and SENTIMENT LOW (negative $Sentiment_{i,t}$).

Research model

To test our hypothesis, we specify the following regression model where the test variable for auditor change ($CHANGE$) and other control variables are included.

$$LNAF_{i,t} = \alpha_1 + \beta_1 CHANGE_{i,t} + \beta_2 SIZE_{i,t} + \beta_3 ROA_{i,t} + \beta_4 LEV_{i,t} + \beta_5 BM_{i,t} + \beta_6 INVREC_{i,t} + \beta_7 QUICK_{i,t} + \beta_8 NEWFIN_{i,t} + \beta_9 LOSS_{i,t} + \beta_{10} BIG4_{i,t} + \beta_{11} SPEC_{i,t} + \beta_{12} OPIN_{i,t} + \beta_{13} ACCLAG_{i,t} + \beta_{14} KOSDAQ_{i,t} + \beta_{15} DSG_{i,t} + YEAR + IND + \varepsilon_{i,t} \quad (7)$$

Variables are defined in Appendix A.

Our dependent variable, $LNAF$, is measured as the natural logarithm of total audit fees in Korean won (KRW). The coefficient on $CHANGE$ (β_1) captures the effect of the auditor change on the auditor's audit pricing. If initial auditors discount audit fees in high sentiment, we expect β_1 to be negative in high sentiment sub-sample.⁵

We control for a number of known determinants of audit fees in the prior literature, which reflect size, complexity, risk, and other client characteristics (e.g. Hay, Knechel, and Wong 2006; DeFond and Zhang 2014). We include client size measured by the natural logarithm of total assets ($SIZE$), the book-to-market ratio (BM), and the sum of inventory and accounts receivable ($INVREC$) for capturing a firms' complexity. Client-specific audit risks are controlled by including leverage ratio (LEV), short-term liquidity ($QUICK$), profitability (ROA , $LOSS$), financing activities ($NEWFIN$), and previous year's accruals component ($ACCLAG$) (Simunic 1980; Francis 1984; Gul and Goodwin 2010). We also include auditor-related variables, which are indicator variables for Big 4 auditors ($BIG4$), industry specialist auditors ($SPEC$), and non-qualified audit opinion ($OPIN$) (Bae, Choi, and Lee 2018). Additionally, the model includes indicator variables for KOSDAQ-listed firms ($KOSDAQ$) and designated firms (DSG) to capture the characteristics of Korean market. Finally, we include year fixed effects and industry fixed effects based on two-digit Korean SIC. Note that all of our analysed results are based on individual firm level clustered standard errors (Peterson 2009), and all continuous variables are winsorized at 1% for both tails.

⁵This study examines the effect of high sentiment on initial discounts by estimating Equation (7) for the high sentiment sub-sample. However, the empirical results of the full sample (high and low) and the low sentiment sub-sample are also presented in Table 3 for comparison.

Sample

The initial sample starts from all KRX listed firms in non-financial industries for the period from 2003 to 2017. We exclude firm-years with non-December fiscal year-ends or with missing data for the construction of the variables. As presented in Panel A of Table 1, the final sample is 17,583 firm-year observations for audit fee analyses and 17,469 firm-year observations for audit hour analyses.⁶ We obtain financial data from FnGuide and audit data (e.g. audit fees, audit hours) from TS2000, both are main Korean financial data providers. Because TS2000 provide audit data from 2003, our sample starts in 2003. Panel B of Table 2 presents the sample distribution by year. 9,084 firm-years are the sub-sample of high investor sentiment and 8,499 are that of low investor sentiment. By yearly, the number of available observations increased with time, from a low of 867 in 2003 to a high of 1,433 in 2013,

which are consistent with Bae, Choi, and Lee (2018). Sub-sample composition between high investor sentiment and low investor sentiment is presented in various forms by year, most firms are included in the high sentiment sub-sample in 2005, and in 2017, most companies are included in the low sentiment sub-sample. Panel C of Table 1 shows the comparison of auditor change and non-change firms by investor sentiment. The total auditor change sample is 3,134 and the observation for auditor change in high sentiment is 1,624, which is slightly more than the comparator sample.

IV. Empirical results

Descriptive statistics and correlations

Panel A of Table 2 reports the descriptive statistics of the variables for the full sample (both high and low sentiment). The mean (median) value of *LNAF*

Table 1. Sample composition.

Panel A: Sample Selection Procedure			
Observations available for FnGuide and TS2000 (2003–2017)			□ 22,881
Less:			
	Firms with non-December fiscal year end		(288)
	Firms with missing information for main variables (Asset, Audit Fees, Sentiment)		(3,199)
	Firms with missing information for other variables		(1,811)
Observations for audit fee analyses			17,583
Less:	Firms with missing audit hour information		(114)
Observations for audit hour analyses			17,469
Panel B: Sample Distribution by Year			
YEAR	SENTIMENT LOW	SENTIMENT HIGH	WHOLE
2003	465	402	867
2004	463	453	916
2005	80	891	971
2006	424	593	1,017
2007	294	780	1,074
2008	640	488	1,128
2009	234	932	1,166
2010	603	674	1,277
2011	565	799	1,364
2012	627	791	1,418
2013	806	627	1,433
2014	873	534	1,407
2015	693	609	1,302
2016	870	324	1,194
2017	862	187	1,049
Total	8,499	9,084	17,583
Panel C: Sample Distribution of Auditor Change by Investor Sentiment			
	SENTIMENT LOW	SENTIMENT HIGH	WHOLE
CHANGE = 0	6,989	7,460	14,449
CHANGE = 1	1,510	1,624	3,134
Total	8,499	9,084	17,583

⁶The Korean public stock market is divided into two main categories: Korea Stock Price Index (KOSPI) and Korea Securities Dealers Automate Quotation (KOSDAQ). The KOSDAQ contains small- and mid-sized firms, and the KOSPI contains relatively larger-sized firms based on more complex listing requirements.

Table 2. Descriptive statistics and correlations.

Panel A: Descriptive statistics

Variable	Mean	S.D.	10%	25%	50%	75%	90%
LNAF	11.119	0.686	10.434	10.714	11.002	11.396	11.983
LNAH	6.669	0.871	5.930	6.246	6.627	7.056	7.608
FEE(KRW)	94,520,430	159,098,500	34,000,000	45,000,000	60,000,000	89,000,000	160,000,000
HOUR	1,188.204	2,130.562	360	512	750	1,156	2,009
SENTIMENT	0.004	0.316	−0.393	−0.193	0.013	0.215	0.400
CHANGE	0.178	0.383	0.000	0.000	0.000	0.000	1.000
SIZE	18.897	1.421	17.356	17.917	18.627	19.594	20.802
ROA	0.035	0.080	−0.052	0.007	0.039	0.076	0.122
LEV	0.436	0.198	0.164	0.277	0.440	0.586	0.696
BM	−0.024	0.771	−1.049	−0.511	0.031	0.506	0.925
INVREC	0.298	0.154	0.103	0.184	0.285	0.397	0.509
QUICK	2.261	2.273	0.707	1.009	1.497	2.509	4.648
NEWFIN	0.104	0.305	0.000	0.000	0.000	0.000	1.000
LOSS	0.270	0.444	0.000	0.000	0.000	1.000	1.000
BIG4	0.506	0.500	0.000	0.000	1.000	1.000	1.000
SPEC	0.187	0.390	0.000	0.000	0.000	0.000	1.000
OPIN	0.004	0.063	0.000	0.000	0.000	0.000	0.000
ACCLAG	−0.032	0.107	−0.137	−0.071	−0.023	0.021	0.072
KOSDAQ	0.579	0.494	0.000	0.000	1.000	1.000	1.000
DSG	0.017	0.131	0.000	0.000	0.000	0.000	0.000

Panel B: Correlation Matrix

	LNAF	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1.LNAH	0.73																
2.SENTIMENT	0.04	0.00															
3.CHANGE	−0.03	−0.02	0.00														
4.SIZE	0.77	0.66	0.10	−0.04													
5.ROA	0.07	0.06	0.29	−0.03	0.21												
6.LEV	0.27	0.20	−0.09	0.03	0.26	−0.21											
7.BM	−0.05	−0.04	−0.19	−0.02	0.15	0.00	−0.05										
8.INVREC	−0.15	−0.14	0.02	−0.01	−0.20	0.06	0.14	0.01									
9.QUICK	−0.23	−0.19	0.02	−0.02	−0.26	0.09	−0.66	−0.05	−0.08								
10.NEWFIN	−0.01	−0.02	−0.10	0.04	−0.12	−0.20	0.07	−0.18	−0.05	−0.02							
11.LOSS	−0.04	−0.03	−0.26	0.03	−0.18	−0.63	0.24	−0.10	−0.05	−0.10	0.22						
12.BIG4	0.38	0.40	0.08	−0.05	0.37	0.09	0.07	0.01	−0.07	−0.10	−0.10	−0.08					
13.SPEC	0.29	0.29	0.07	−0.03	0.28	0.06	0.06	−0.02	−0.12	−0.07	−0.05	−0.04	0.42				
14.OPIN	0.03	0.00	−0.06	0.02	−0.03	−0.10	0.04	−0.05	−0.01	−0.01	0.10	0.07	−0.03	−0.02			
15.ACCLAG	−0.02	0.00	0.06	−0.06	0.10	0.17	−0.08	0.12	0.12	0.05	−0.13	−0.15	0.03	−0.01	−0.02		
16.KOSDAQ	−0.38	−0.31	−0.04	0.00	−0.52	−0.07	−0.13	−0.23	0.01	0.17	0.13	0.11	−0.24	−0.16	0.01	−0.06	
17.DSG	0.09	0.04	−0.02	0.08	0.01	−0.03	0.05	−0.04	−0.01	−0.03	0.05	0.03	0.02	0.02	0.03	−0.03	0.00

Definition of variables is as follows. *LNAF* is the natural logarithm number of total audit fees; *LNAH* is the natural logarithm number of total audit hours; *SENTIMENT* is individual firm-level investor sentiment measure calculated based on the first principle component on five proxies; *CHANGE* a dummy variable that takes 1 if auditor of the firm is changed for the period, otherwise 0; *SIZE* is the natural logarithm number of total assets; *ROA* is operating income scaled by total asset; *LEV* is total liabilities divided by total assets; *BM* is book value of equity divided by market value of the firm; *INVREC* is sum of inventory and account receivable scaled by total asset; *QUICK* is current liability divided by current asset; *NEWFIN* is a dummy variable that takes 1 if the firm increase the book value of equity during period, otherwise 0; *LOSS* is a dummy variable that takes 1 if the firm reported a net loss, otherwise 0; *BIG4* is a dummy variable that takes 1 if the firm is audited by Big 4 auditors (PwC, KPMG, Deloitte, and EY), otherwise 0; *SPEC* is a measure of industry specialist auditors defined as a dummy variable takes 1 if auditor has a portion over 30% within the industry and has a gap with a following auditor over 10%, based on client total sales. The market share is calculated as the sum of the auditor's client total sales for each industry-year divided by the sum of all auditors' client total sales in the industry-year; *OPIN* is a dummy variable that takes 1 if the firm receive audit opinion except for qualified opinion, otherwise 0; *ACCLAG* is lagged total accruals divided by lagged total asset, accruals is calculated by subtracting operating cash flow from net income; *KOSDAQ* is a dummy variable that takes 1 if the firm is listed in KOSDAQ market, otherwise 0; *DSG* is a control for designation in Korean audit market, a dummy variable that takes 1 if total audit fee is more than doubled over previous period, otherwise 0.

Refer to Table 2 for variable definitions. **BOLD** represents significance at 1% (two-tailed).

is 11.119 (11.002) and the mean (median) value of *LNAH* is 6.669 (6.627). The mean values of audit fees (*FEE*) and audit hours (*HOURL*) are 94,520,430 KRW and 1,188.204 KRW, respectively. The mean value of *SENTIMENT* is close to 0 (0.004) because it is standardized to eliminate market factors. The mean value of *CHANGE* is 0.178, which indicates that about 17.8% of the sample changes auditors.

BIG4 and *SPEC* account for approximately 50.6% and 18.7%, respectively. The other variables are generally consistent with prior research.

The Pearson correlation matrix is presented in Panel B of Table 2, and the results are consistent with prior research. For example, audit fees (*LNAF*) have a positive relationship with audit hours (*LNAH*) but not the same variable. Audit fees (*LNAF*) are

positively correlated with the risk of audited firms such as firm size (*SIZE*), profitability (*ROA*, *LOSS*), and leverage (*LEV*). Audit fees (*LNAF*) are also positively associated with auditors' reputations and abilities proxied by Big4 auditors (*BIG4*) and industry specialist auditors (*SPEC*). Individual firm-level sentiment measure (*SENTIMENT*) is slightly positively correlated with audit fees (*LANF*), but not with audit hours (*LNAH*) significantly.

High investor sentiment and initial engagement discounts

Table 3 reports the empirical results regarding the initial discounts. Columns 1, 2, and 3 are the results of the full sample (*WHOLE*), the low sentiment sub-sample (*SENTIMENT LOW*, *negative sentiment*), and the high sentiment sub-sample (*SENTIMENT HIGH*, *positive sentiment*), respectively.

In column 1, the coefficient on *CHANGE* is significantly negative, supporting the existence of initial engagement discounts consistent with previous studies (e.g. Rho, Bae, and Cho 2004; Lee, Park, and Jeong 2011; Desir, Casterella, and Kokina 2014; Jeong and Ki 2016). The results in columns 2 and 3 show the effect of investor sentiment on initial discounts. Specifically, the coefficient on *CHANGE* is not significant in *SENTIMENT LOW*, while it is significantly negative in *SENTIMENT HIGH*. In addition, the coefficient in *SENTIMENT HIGH* is larger and more significant than in *WHOLE*, indicating that auditors discount initial audit fees in high sentiment periods and that investor sentiment influences audit risk assessments other than litigation risk. Considering the lower audit fees and relatively less litigious environment in Korea, the results suggest that auditors perceive the overall audit risk including reputation loss and regulation intervention as low during optimistic mood periods, decreasing the possibility that auditors closely monitor managerial opportunistic behaviour in high sentiment times. It also shows that investor sentiment surrounding a firm can affect auditors' mood state, even if it does not seem to be directly related to auditors. The economic effect of auditor change in high investor sentiment is approximately a 5.7% decrease in audit fees.

Table 3. High investor sentiment and initial engagement discounts.

<i>DV = LNAF</i>	<i>WHOLE</i> (1)	<i>SENTIMENT LOW</i> (2)	<i>SENTIMENT HIGH</i> (3)
<i>Intercept</i>	3.392***	3.835***	2.991***
<i>CHANGE</i>	-0.025***	0.005	-0.057***
<i>SIZE</i>	0.386***	0.366***	0.405***
<i>ROA</i>	-0.471***	-0.456***	-0.370***
<i>LEV</i>	-0.000	0.018	-0.037
<i>BM</i>	-0.116***	-0.131***	-0.101***
<i>INVREC</i>	0.163***	0.126**	0.203***
<i>QUICK</i>	-0.005	-0.005	-0.006
<i>NEWFIN</i>	0.062***	0.090***	0.020
<i>LOSS</i>	0.052***	0.050***	0.049***
<i>BIG4</i>	0.124***	0.143***	0.108***
<i>SPEC</i>	0.050***	0.049**	0.052***
<i>OPIN</i>	0.415***	0.385**	0.435**
<i>ACCLAG</i>	-0.362***	-0.324***	-0.358***
<i>KOSDAQ</i>	-0.028	-0.038*	-0.012
<i>DSG</i>	0.357***	0.370***	0.323***
<i>Year fixed</i>	Yes	Yes	Yes
<i>Industry fixed</i>	Yes	Yes	Yes
<i>Number of Obs.</i>	17,583	8,499	9,084
<i>Adjusted R-sq.</i>	0.679	0.631	0.721

Refer to Table 2 for variable definitions. Industry fixed effects based on two-digit. *t*-statistics are calculated with individual firm-level clustered standard errors. *, ** and *** represent significance at 10%, 5% and 1%, respectively (two-tailed).

In general, the coefficients of the control variables are consistent with prior literature. Audit fees are significantly associated with client firms size, complexity of audit work, and client-specific risks (*SIZE*, *INVREC*, *BM*, *NEWFIN*, *OPIN*). The profitability of client firms is also negatively associated with audit fees (*ROA*, *LOSS*). The positive coefficients of *BIG4* and *SPEC* are consistent with the fee premium for specialized auditors with high skills and knowledge for audits (Bae, Choi, and Lee 2018).

High investor sentiment, initial engagement discounts, and auditor characteristics

The literature shows that Big4 auditors (*BIG4*) and industry specialist auditors (*SPEC*) charge higher audit fees for their greater ability and additional efforts to mitigate potential litigation risk (DeFond and Zhang 2014). Given the substantial influence of these auditors on the audit market, it is important to examine whether the effect of high investor sentiment on initial discounts is maintained even if the auditor is changed to *BIG4* or *SPEC*. The results are provided in Table 4. Columns (1) and (2) show the results of *BIG4* and *SPEC*, respectively.

Table 4. High investor sentiment, initial engagement discounts, and auditor characteristics.

DV = LNAF	SENTIMENT HIGH (1)	SENTIMENT HIGH (2)
Intercept	2.983***	2.980***
CHANGE	−0.031**	−0.041***
CHANGE X BIG4	−0.051***	
CHANGE X SPEC		−0.083**
BIG4	0.117***	0.107***
SPEC	0.051***	0.066***
Controls	Yes	Yes
Year fixed	Yes	Yes
Industry fixed	Yes	Yes
Number of Obs.	9,084	9,084
Adjusted R-sq.	0.721	0.721

Refer to Table 2 for variable definitions. Industry fixed effects based on two-digit. *t*-statistics are calculated with individual firm-level clustered standard errors. *, ** and *** represent significance at 10%, 5% and 1%, respectively (two-tailed).

The coefficients of columns (1) and (2) on *CHANGE* are all significantly negative, indicating the existence of initial discounts. The coefficients of *CHANGE* × *BIG4* and *CHANGE* × *SPEC*, which are the incremental effects of *BIG4* and *SPEC*, are significant and negative. This shows that more initial discounts are offered to clients audited by *BIG4* and *SPEC* in high investor sentiment. Given the high risk sensitivity of *BIG4* and *SPEC*, the results suggest that its impact is greater for auditors with greater litigation risk. Their fee premiums, shown by the significantly positive coefficients of *BIG4* and *SPEC*, may enable them to discount their initial audit fees.

High investor sentiment and audit hours of initial engagement auditors

According to the literature, audit hours are considered to directly capture audit effort (DeFond and Zhang 2014). If auditors discount initial audit fees without decreasing audit hours, the negative effect of initial discounts on audit quality is less likely. To examine this, we investigate the association between auditor change and audit hours in high investor sentiment and report the results in Table 5. Column (1) shows the result for the high sentiment sample, and columns (2) and (3) show the results considering *BIG4* and *SPEC*, respectively.

In column (1), the coefficient on *CHANGE* is not significant, indicating that the initial discounting behaviour in column (3) of Table 3 is not driven by reduced audit hours. The coefficients on

Table 5. High investor sentiment and audit hours of initial engagement auditors.

DV = LNAH	SENTIMENT HIGH (1)	SENTIMENT HIGH (2)	SENTIMENT HIGH (3)
Intercept	−1.279***	−1.293***	−1.299***
CHANGE	−0.006	0.044*	0.025
CHANGE X BIG4		−0.097***	
CHANGE X SPEC			−0.158***
BIG4	0.263***	0.281***	0.262***
SPEC	0.178***	0.176***	0.204***
Controls	Yes	Yes	Yes
Year fixed	Yes	Yes	Yes
Industry fixed	Yes	Yes	Yes
Number of Obs.	9,023	9,023	9,023
Overall R-sq.	0.550	0.551	0.551

Refer to Table 2 for variable definitions. Industry fixed effects based on two-digit. *t*-statistics are calculated with individual firm-level clustered standard errors. *, ** and *** represent significance at 10%, 5% and 1%, respectively (two-tailed).

CHANGE in columns (2) and (3) are not negative, implying that auditor change to non-Big4 and non-specialist do not decrease audit hours in high investor sentiment periods on average. However, the incremental effects of *BIG4* and *SPEC* are significantly negative when investor sentiment is high. This suggests that reduced audit hours contribute to initial engagement discounts when auditors are changed to *BIG4* or *SPEC*. The results do not mean that the decrease in audit hours of *BIG4* and *SPEC* results in audit failure because the finding of prior studies, more audit hours of *BIG4* and *SPEC*, may enable *BIG4* and *SPEC* to reduce audit effort to the extent that audit quality is not impaired (DeFond and Zhang 2014). Less audit hours of *BIG4* and *SPEC* may be the result of auditors' expectations that investors are generous to financial statement errors in high sentiment periods.

Additional analyses

To ensure the robustness of our results, we conduct several additional analyses. First, we consider the types of auditor change such as Big 4 to Big 4, Big 4 to non-Big 4, non-Big 4 to Big 4, and non-Big 4 to non-Big 4. Table 6 presents the results.

As shown in Table 6, the coefficients of auditor change are consistent with our main findings except for non-Big4 to non-Big4 (*NBIG4_NBIG4*). Specifically, the coefficients of *BIG4_NBIG4*, *BIG4_BIG4*, and *NBIG4_BIG4* are negatively significant. Even if Craswell and Francis (1999) provide evidence that initial discounts exist in auditor

Table 6. Additional analysis: Types of auditor change.

DV = LNAF	SENTIMENT HIGH (1)
Intercept	2.984***
BIG4_NBIG4	−0.068***
BIG4_BIG4	−0.064***
NBIG4_NBIG4	−0.006
NBIG4_BIG4	−0.105***
Controls	Yes
Year fixed	Yes
Industry fixed	Yes
Number of Obs.	9,084
Adjusted R-sq.	0.721

Refer to Table 2 for variable definitions. Industry fixed effects based on two-digit. *t*-statistics are calculated with individual firm-level clustered standard errors. *, ** and *** represent significance at 10%, 5% and 1%, respectively (two-tailed).

change from non-Big4 to Big4, the significantly negative coefficient of *NBIG4_BIG4* is somewhat surprising because audit fees charged by Big4 are generally higher than non-Big4. This can be interpreted as indicating that high investor sentiment substantially influences initial audit pricing.

Second, we use interaction terms and re-examine the effect of high investor sentiment on initial discounts for the whole sample. The results are presented in Table 7. Columns (1) and (2) use the dummy variable *SENTIMENT_HIGH* and the continuous variable *SENTIMENT*, respectively. The variables of interest are *SENTIMENT_HIGH* × *CHANGE* and *SENTIMENT* × *CHANGE* in columns (1) and (2), respectively.

The coefficients of *SENTIMENT_HIGH* × *CHANGE* and *SENTIMENT* × *CHANGE* are significant and negative, consistent with our main results. This increases the robustness of our findings by confirming that our main results are maintained in the full sample with low and high investor sentiment observations. Meanwhile, Table 7 shows that *SENTIMENT_HIGH* (*SENTIMENT*) is negatively associated with *LNAF*, indicating that investor sentiment is an important factor affecting audit fees.

V. Conclusion

This study examines whether high sentiment affects auditors' risk assessment through initial audit discounts in Korea with less litigious environment. The prior finding that audit fees are lower in high sentiment is interpreted as the reduction of

Table 7. Additional analysis: Interaction model.

DV = LNAF	WHOLE (1)	WHOLE (2)
Intercept	3.380***	3.349***
CHANGE	0.009	−0.026***
SENTIMENT_HIGH X CHANGE	−0.065***	
SENTIMENT_HIGH	−0.018**	
SENTIMENT X CHANGE		−0.122***
SENTIMENT		−0.055***
Controls	Yes	Yes
Year fixed	Yes	Yes
Industry fixed	Yes	Yes
Number of Obs.	17,583	17,583
Adjusted R-sq.	0.679	0.680

Refer to Table 2 for variable definitions. Industry fixed effects based on two-digit. *t*-statistics are calculated with individual firm-level clustered standard errors. *, ** and *** represent significance at 10%, 5% and 1%, respectively (two-tailed).

litigation risk (Amin, Eshleman, and Guo 2021a), but the legal environment is not the same for each country; thus, it is necessary to investigate whether the result is consistent in a weak legal system to generalize it.

Using Korean public firm-years from 2003 to 2017, we find that initial discounts exist in high investor sentiment, but not in low investor sentiment. Further, the results are maintained, and the initial audit hours are also low when auditors are changed to Big4 or specialists. In addition, analyses considering the auditor change type show that our main results are generally consistent. Our findings suggest that high sentiment makes auditors judge engagement risk less conservatively, increasing that they are unlikely to detect firms' opportunistic behaviour when investor sentiment is high.

This study has the following limitations. First, we measure the investor sentiment variable following prior studies (e.g. Ryu, Kim, and Yang 2017), but it is a modified version of Baker and Wurgler (2006) which is appropriate for the Korean environment. Therefore, our findings may not be comparable to other studies measuring the variable based on Baker and Wurgler (2006).

Second, individual investors account for two-thirds of the Korean stock market and have speculative and behaviourally biased characteristics (e.g. Chung, Park, and Ryu 2016). Therefore, our results may not be generalizable to other countries whose individual investors and stock market are characterized differently.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix A

Variables	Definition
LNAF	the natural logarithm number of total audit fees
LNAH	the natural logarithm number of total audit hours
SENTIMENT	individual firm-level investor sentiment measure calculated based on the first principle component on five proxies
CHANGE	a dummy variable that takes 1 if auditor of the firm is changed for the period, otherwise 0
SIZE	the natural logarithm number of total assets
ROA	operating income scaled by total asset
LEV	total liabilities divided by total assets
BM	book value of equity divided by market value of the firm
INVREC	sum of inventory and account receivable scaled by total asset
QUICK	current liability divided by current asset
NEWFIN	a dummy variable that takes 1 if the firm increase the book value of equity during period, otherwise 0
LOSS	a dummy variable that takes 1 if the firm reported a net loss, otherwise 0
BIG4	a dummy variable that takes 1 if the firm is audited by Big 4 auditors (PwC, KPMG, Deloitte, and EY), otherwise 0
SPEC	a measure of industry specialist auditors defined as a dummy variable takes 1 if auditor has a portion over 30% within the industry and has a gap with a following auditor over 10%, based on client total sales. The market share is calculated as the sum of the auditor's client total sales for each industry-year divided by the sum of all auditors' client total sales in the industry-year.
OPIN	a dummy variable that takes 1 if the firm receive audit opinion except for qualified opinion, otherwise 0
ACCLAG	lagged total accruals divided by lagged total asset, accruals is calculated by subtracting operating cash flow from net income
KOSDAQ	a dummy variable that takes 1 if the firm is listed in KOSDAQ market, otherwise 0
DSG	a control for designation in Korean audit market, a dummy variable that takes 1 if total audit fee is more than doubled over previous period, otherwise 0